Machine Learning and Data Mining

- 15CSE402

Group-1

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1. Introduction

Usually, individuals consume a few items from a large range of potential items. However, there are several circumstances where it is important to make predictions for new items based on previously consumed items for a person, rather than just recommending fresh items. In their ability to capture important information in historical behavior, matrix factorisation is a commonly used technique.

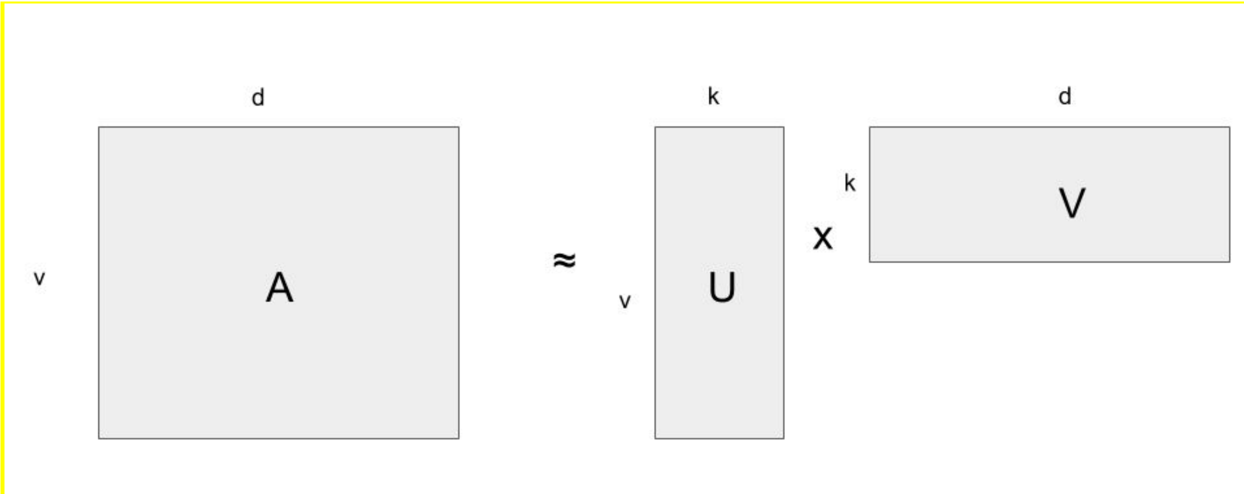
2. Objective

To build a model to predict user-song compatibility using matrix factorisation and cosine similarity methods and is analyzed in terms of loss calculation.

3. Problem statement and Assumption

Predicting Consumption Patterns with Repeated and Novel Events

4. Architecture



Matrix factorization algorithms work by decomposing the original matrix into two matrices one is the upper triangle (U), and the other is the lower triangle (L).

In matrix factorization, we basically break a matrix into usually 2 smaller matrices each with smaller dimensions. These matrices are often called 'Embeddings'. We can have variants of Matrix Factorization-> 'Low Rank MF' , 'Non-Negative MF' (NMF) and so on.

Here, 'Low Rank Matrix Factorization' is being used.Embeddings are created for both users as well as the item; song in our case. The number of dimensions or the so called 'Latent Factors' in the embeddings is a hyperparameter to deal with in this implementation of Collaborative Filtering.

5. Related work

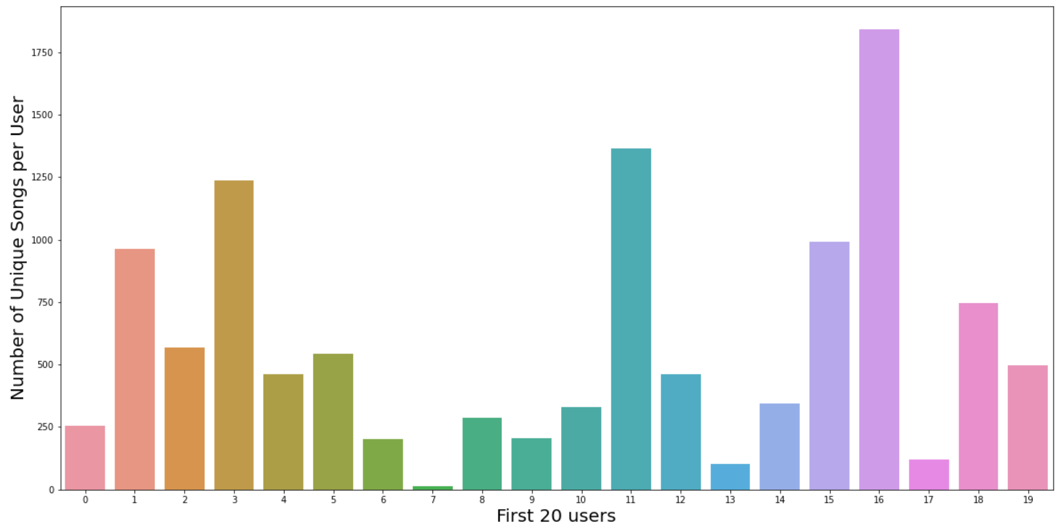
Matrix factorization techniques are widely used in research on recommender systems. In such applications, the model is primarily tasked with learning user preferences from historical observations and predicting future user-item interactions with a focus on the discovery of new items [5], [6], [7].

6. Data collection and preparation

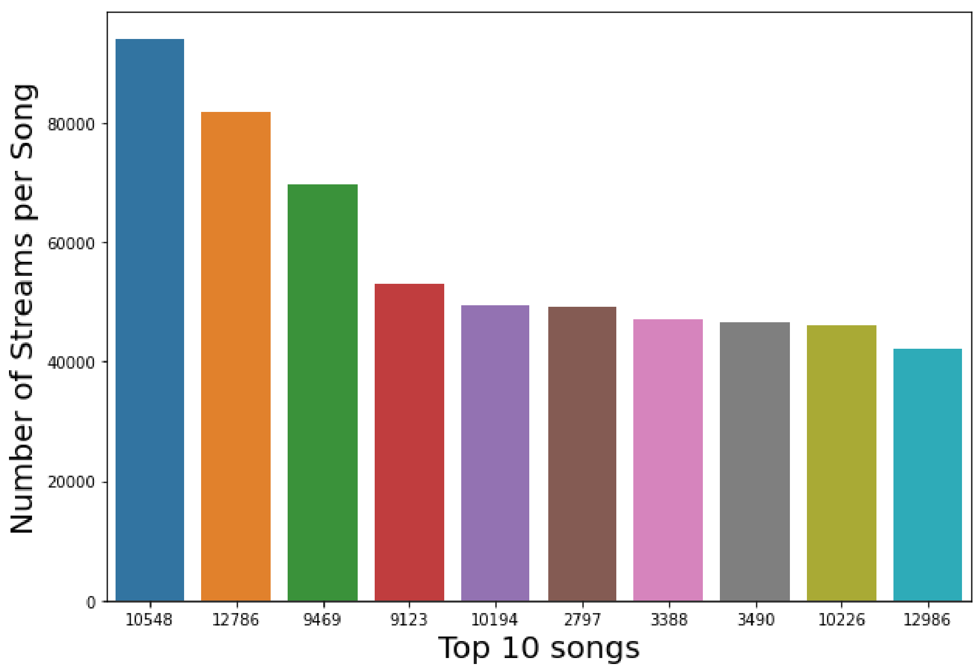
a. Feature Selection and/or Feature Extraction

We also computed the correlation,since there are limited features in the dataset, feature extraction does not seem feasible.

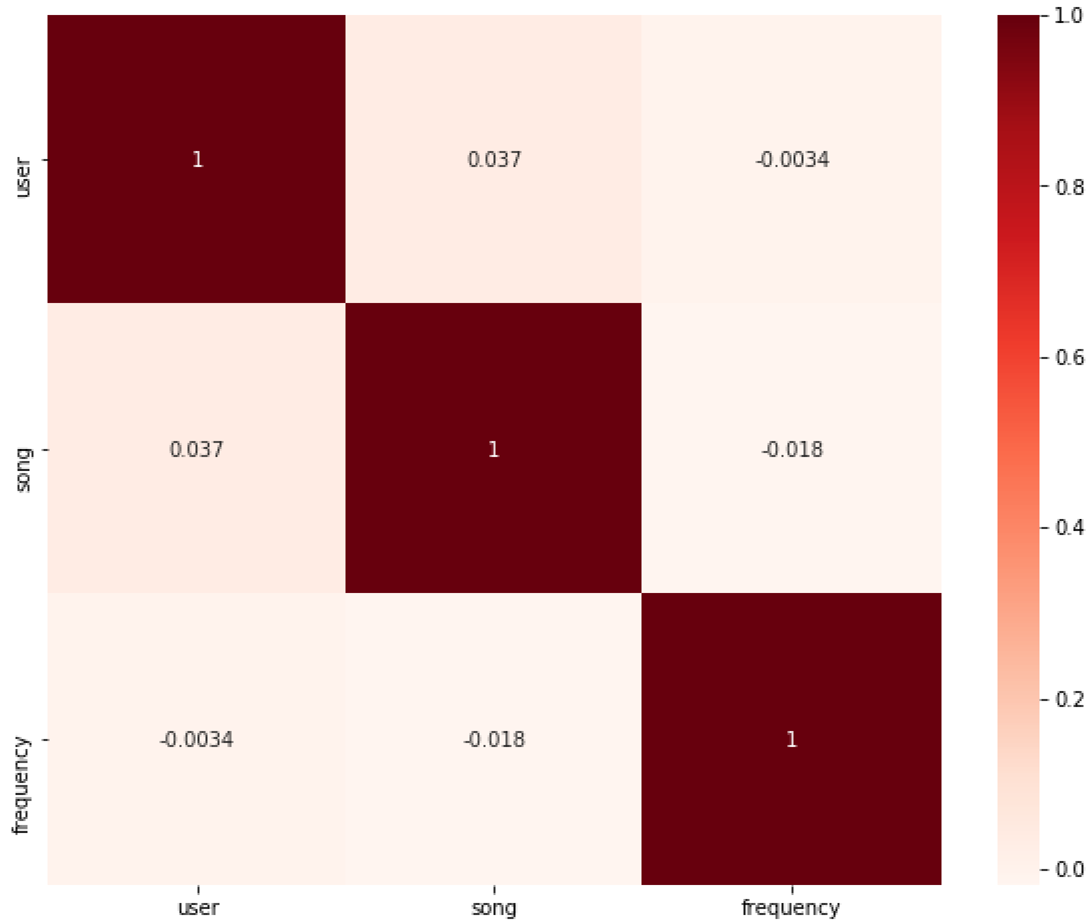
b. Data Visualization and Hypothesis



The above plot is bar plot where we can see the users with different mindsets - from user listening to selected number of songs to users listening to wide variety of songs



Here we can see the top 10 songs streamed by users



Here we can see the correlation between each pair of columns.

c. Data Cleaning & Pre-Processing

We used a clean dataset that had columns(user,song,frequency). No preprocessing was required for the dataset

7. Data Modeling and Inference

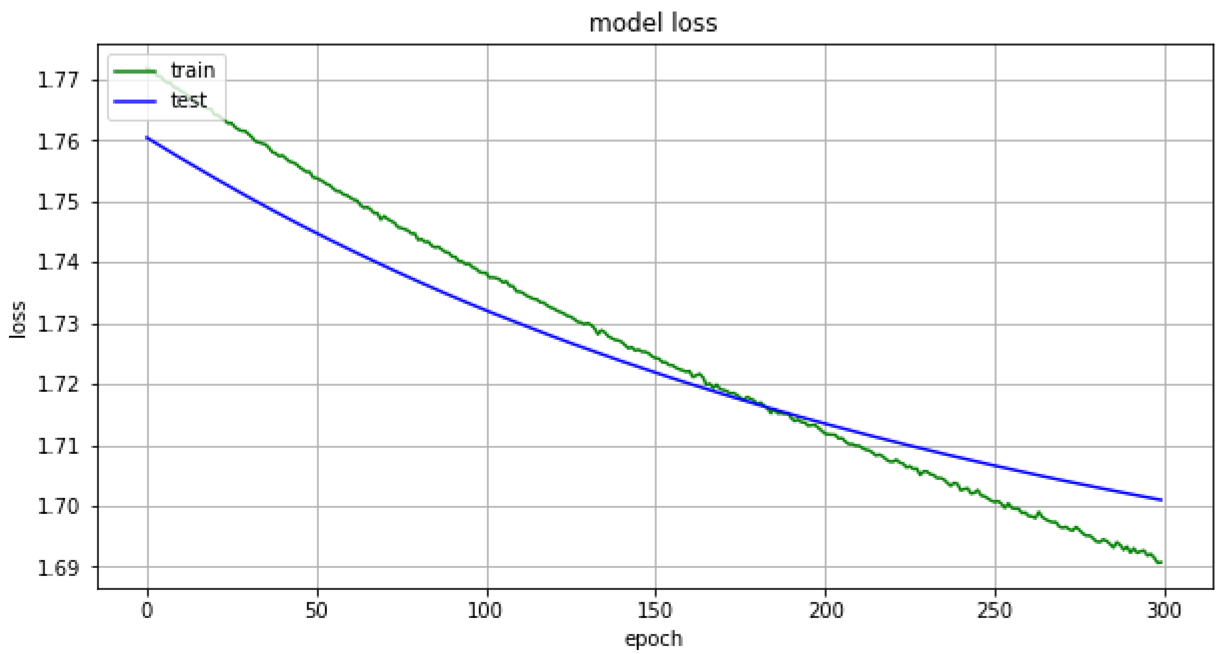
a. Model

For the first model embeddings were used for each user and song using embeddings function from keras.layers and later embeddings were flattened from matrix form to 1-d array. Later dot product of both user and song vectors is calculated and created a model using it. We used Adam optimiser to optimize our model. Then model is fitted and was run for 300 epochs and plotted the graph between training loss and testing loss

For the second model cosine similarity on both users and songs is taken. Through this model prediction of similar songs to a particular song, prediction of similar users(like similar mindsets). A function which predicts the frequency given user and song.MAE between actual and predicted values for few users is calculated

b. Performance Evaluation and result discussion

The loss has been computed with respect to every epoch and a graph has been plotted about the same shown below. We can see that as epochs increase the training loss and validation loss tends to converge. We observe different epoch vs loss graphs with changes in dropout value and hyper parameter value.



b. Model Tweaking, Regularization, Hyper Parameter Tuning

We have used the Dropout function to avoid overfitting of the model. In our model when we are converting user and song inputs into embeddings with the help of keras. We have a hyperparameter called n_latent_factors which helps us in determining the dimensions of our embeddings. The values were tuned so as to reduce the difference between training loss and validation loss.

8. Conclusion and Future Enhancements

Using a real-world dataset, the user-song compatibility was computed using matrix factorisation and the loss was analysed and cosine similarity was computed to group the similar songs or users or predict frequencies.

In future, we will try making the existing model more efficient or making a new model with better efficiency.

9. References:

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[2] Liu, Yue et al. "Characterizing and Predicting Repeat Food Consumption Behavior for Just-in-Time Interventions". *Proceedings of the 9th International Conference on Digital Public Health - DPH2019*. (2019).

[3]Xiwang Yang, Yang Guo, Yong Liu, & Harald Steck (2014). A survey of collaborative filtering based social recommender systems*Computer Communications, 41*, 1 - 10.

[4]X. Ning and G. Karypis, "SLIM: Sparse Linear Methods for Top-N Recommender Systems," *2011 IEEE 11th International Conference on Data Mining*, Vancouver,BC,

[5] X. Yang, Y. Guo, Y. Liu, and H. Steck, “*A survey of collaborative filtering based social recommender systems,*” Comput. Commun., vol. 41, pp. 1–10, Mar. 2012011, pp. 497-506, doi: 10.1109/ICDM.2011.134.

[6]] X. Ning and G. Karypis, “Slim: Sparse linear methods for top-n recommender systems,” *in Proceedings of the 11th IEEE International Conference on Data Mining*, ser. ICDM ’11. IEEE, 2011, pp. 497–506.

[7] D. Kotkov, S. Wang, and J. Veijalainen, “A survey of serendipity in recommender systems,” *Know.-Based Syst.*, vol. 111, no. C, pp. 180–192, Nov. 2016